PROBLEM DA5

Power of a Wind Turbine

Original statement · DA5.pdf

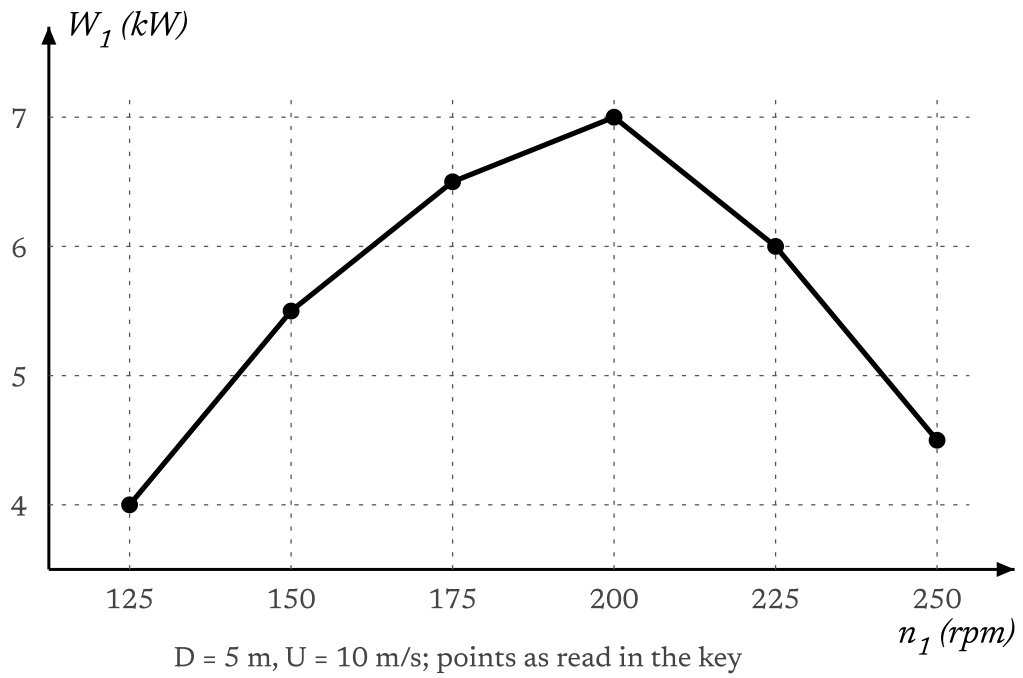
Π theorem | Re ≫ 1 | Experimental curve

OFFICIAL PROBLEM STATEMENT

We want to investigate the performance of a wind turbine of given geometrical shape and diameter D rotating with angular velocity n subject to a wind stream of velocity U. Assume in the development that the velocity is sufficiently small compared with that of sound that air effectively behaves as an incompressible fluid of constant density and kinematic viscosity ρ and ν.

- Use the dimensional analysis to reduce the parametric dependence of the power produced by the wind turbine W.
- Simplify the results by assuming that the effect of viscosity μ is negligible. Justify the simplification.

We have measured the variation with n of the power delivered by a given turbine of diameter $D = 5$ m on a day when the wind was blowing at $U = 10$ m/s, given the results shown in the plot below.



D = 5 m, U = 10 m/s; points as read in the key

Statement figure (redrawn from DA5.pdf): power W_1 delivered by the turbine of diameter D = 5 m against its angular velocity n_1 for a wind of U = 10 m/s; the dots are the values read in the official key (4, 5.5, 6.5, 7, 6 and 4.5 kW at 125 to 250 rpm), a reading of the plot

Based on these results, address the following three questions:

- If the wind velocity on a different day is $U = 15$ m/s, determine the angular velocity required for the turbine to deliver maximum power as well as the resulting power.
- If we fix the angular velocity at $n = 200$ rpm, obtain the power response as a function of the wind velocity $W(U)$.
- Consider a geometrically similar turbine of diameter $D = 10$ m. Obtain the power that it would deliver if placed rotating at $n = 175$ rpm in a wind stream of velocity $U = 20$ m/s.

The sheet prints the last diameter as "10.^{m}" (the unit slipped into a superscript); it is $D = 10$ m.

`Source: DA5.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; limits, Eq. 6.55; physical similarity, Eqs. 6.57-6.59.`

Worked solution

Hypotheses, Data and Dimensional Table

- Incompressible air (low Mach number, as the statement says): no speed of sound in the list. No free surface: no gravity.
- Measured state, subscript 1: $D_1 = 5$ m, $U_1 = 10$ m/s, and the curve $W_1(n_1)$, read at the points of the key (a reading of the plot).

Parameter	Meaning	Dimensions
W	power delivered (unknown)	ML^2T^{-3}
U	wind velocity	LT^{-1}
n	angular velocity	T^{-1}
D	rotor diameter	L
ρ	air density	ML^{-3}
μ	air viscosity	$ML^{-1}T^{-1}$

Application of the Π Theorem

$$W = W(U, n, D, \rho, \mu)$$

$n = 5$ parameters; (U, D, ρ) have independent dimensions (rows $U(1,0,-1)$, $D(1,0,0)$, $\rho(-3,1,0)$, $\det = -1$, as in DA1), so two groups remain.

Power. $[W] = [\rho]^a[U]^b[D]^c$: $M:\ 1 = a$; $T:\ -3 = -b$; $L:\ 2 = -3 + 3 + c$; so $W/(\rho U^3 D^2)$.

Rotation. $[n] = [U]^b[D]^c$: $T:\ -1 = -b$; $L:\ 0 = b + c$; so $b = 1, c = -1$ and the group is $n/(U/D) = nD/U$, the ratio of the blade-tip speed to the wind speed.

Viscosity. As in DA1, $\mu/(\rho U D)$, the inverse of the Reynolds number.

$$\frac{W}{\rho U^3 D^2} = f\left(\frac{nD}{U}, \frac{\rho U D}{\mu}\right)$$

Negligible Viscosity

In typical applications $\rho U D/\mu \gg 1$: with $U = 10$ m/s, $D = 5$ m and $\nu \simeq 1.5 \times 10^{-5}$ m^2/s for air (a standard value, not given in the statement), $\mathrm{Re} = UD/\nu \simeq 3 \times 10^6$. Viscosity then does not influence the flow through the rotor, and with ρ in the independent set the Reynolds number drops out (Notes.pdf, Eq. 6.55):

$$\frac{W}{\rho U^3 D^2} = f\left(\frac{nD}{U}\right)$$

Two states with the same nD/U are physically similar, and their powers are related by equal $W/(\rho U^3 D^2)$. The air density is the same in all the cases below, so it cancels.

Maximum Power at U = 15 m/s

On the measured curve, the maximum is $W_1 = 7$ kW at $n_1 = 200$ rpm. The state at $U_2 = 15$ m/s similar to it (same D) has

$$\frac{n_1}{U_1/D} = \frac{n_2}{U_2/D} \implies n_2 = \frac{U_2}{U_1}\, n_1 = \frac{15}{10} \times 200 = 300 \text{ rpm}$$

$$\frac{W_1}{\rho U_1^3 D^2} = \frac{W_2}{\rho U_2^3 D^2} \implies W_2 = \left(\frac{U_2}{U_1}\right)^3 W_1 = 1.5^3 \times 7 = 3.375 \times 7 = 23.625 \text{ kW}$$

Since f has a single maximum, the maximum of W at fixed U is reached at the same nD/U as in the test.

Power Curve at n = 200 rpm

Each measured point (n_1, W_1) is similar to the state at $n = 200$ rpm with the wind that keeps nD/U:

$$U = \frac{n}{n_1} U_1 = \frac{200}{n_1} \times 10 \text{ m/s}, \qquad W = \left(\frac{U}{U_1}\right)^3 W_1$$

n_1 (rpm)	W_1 (kW)	U (m/s)	$(U/U_1)^3$	W (kW)
125	4	16	4.096	16.38
150	5.5	13.33	2.370	13.04
175	6.5	11.43	1.493	9.70
200	7	10	1	7
225	6	8.89	0.702	4.21
250	4.5	8	0.512	2.30

Larger Turbine

For $D = 10$ m, $n = 175$ rpm and $U = 20$ m/s, the similar measured point has the same nD/U:

$$n_1 = n \, \frac{D}{D_1} \, \frac{U_1}{U} = 175 \times \frac{10}{5} \times \frac{10}{20} = 175 \text{ rpm} \implies W_1 = 6.5 \text{ kW}$$

$$W = \left(\frac{U}{U_1}\right)^3 \left(\frac{D}{D_1}\right)^2 W_1 = 2^3 \times 2^2 \times 6.5 = 8 \times 4 \times 6.5 = 208 \text{ kW}$$

Answers and Checks

- Reduced law: $W/(\rho U^3 D^2) = f(nD/U, \rho UD/\mu)$, and $f(nD/U)$ for $\text{Re} \gg 1$.
- At $U = 15$ m/s: maximum power at $n = 300$ rpm, $W = 23.6$ kW.
- At $n = 200$ rpm: the table above, from 16.4 kW at 16 m/s to 2.3 kW at 8 m/s.
- Turbine of 10 m at 175 rpm and 20 m/s: $W = 208$ kW.
- Dimensions: $\rho U^3 D^2$ is $ML^{-3} \cdot L^3 T^{-3} \cdot L^2 = ML^2 T^{-3}$; nD/U is $T^{-1} L/(LT^{-1}) = 1$, so rpm can be used on both sides without conversion.
- Check against the official key: DA5.pdf gives $n = 300$ rpm and $W = 23.625$ kW, the table U = 16, 13.3, 11.43, 10, 8.89, 8 m/s with W = 16.38, 13.03, 9.7, 7, 4.2, 2.3 kW, and $W = 208$ kW for the 10 m turbine, the same results (the key truncates $2.370 \times 5.5 = 13.037$ to 13.03; rounded, 13.04).